

Harvard University
Physics 141 Pset 3

Due October 12, 2025

1. A discrete-time random walk in d dimensions starts at the origin and, each time step, takes an independent step. We will simulate random walks on a hypercubic lattice with spacing a . To do so, for each time step Δt , choose one of the directions along the axes of the lattice randomly with equal probability, and move by distance a in that direction. After n steps, the position of the walker is $r_n = \sum_{k=1}^n \Delta r_k$. The mean squared displacement grows linearly with the number of steps n

$$\langle r_n^2 \rangle = n a^2$$

And the diffusion constant can be written as

$$D = \frac{a^2}{2d \Delta t}$$

Assume $a = 1$ and $\Delta t = 1$. Then, $\langle r_n^2 \rangle = n$.

- (a) For dimensions $d = 1, 2, 3$, simulate an ensemble of $N = 1000$ independent random walkers. Use $a = 1$ and $\Delta t = 1$. Simulate for $n = 100$ steps. At each step, choose one of the directions along the lattice with equal probability and move to the neighboring node. For each number of dimensions ($d = 1, 2, 3$), plot the mean squared displacement as a function of step number. On these plots, overlay the theoretical lines $\langle r_n^2 \rangle = n$.
- (b) Introduce a “stay put” probability. What this means is that for each step, there is a probability p that the particle does not move. Perform the same simulations as in (a), except with this “stay put” probability. Use $p = 0.3$. Compare to your results in (a).
- (c) How does introducing a “stay put” probability affect the slope of your mean squared displacement? Fit a curve of form $y = mx$ to your plots and use this curve to estimate D_{eff} , the effective diffusion constant when a “stay put” probability is introduced. What is $\frac{D_{eff}}{D}$? Here, D denotes the diffusion constant when the “stay put” probability is zero ($p = 0$). Can you write your answer in terms of given variables?

We will now simulate only 1D random walks with nonconstant spatial step sizes. In a 1D random walk, the step distance Δx can be drawn from a distribution $p(\Delta x)$. Assume that this distribution has mean $\mu = 0$, is symmetric around zero, and has variance σ^2 . In this case, as you will see, the mean squared displacement grows linearly. Simulate 1D random walks ($N = 1000$, number of time steps $T = 100$, $\Delta t = 1$) with the step Δx being sampled from a Gaussian distribution

$$\Delta x \sim N(0, \sigma^2)$$

- (d) To start, for $\sigma = 1$, plot the mean squared displacement as a function of number of steps. Fit a linear function of form $y = mx$ to this data and estimate the slope. Get rid of the probability of staying put.
 - (e) Repeat (a) for various values of σ . Plot the slope of the line as a function of σ .
 - (f) You should see that the slope of the line as a function of σ is a quadratic equation (in this case, $m = \sigma^2$). Using this result, write a formula for the diffusion constant D as a function of σ .
2. (Bonus) For this problem, we will simulate discrete-time random walks on a 2D square lattice with spacing $a = 1$ and for time step $\Delta t = 1$. We will implement self-avoiding behavior (the walker prefers not to revisit previously visited sites), which should tend to push the walker outwards. To implement self-avoidance, maintain a record of all visited sites. At each time step Δt ,
- i If there exists at least one unvisited neighbor, choose uniformly among the unvisited neighboring lattice points.
 - ii Otherwise (all neighbors have been visited), choose uniformly among all four neighboring lattice points.

This implements a simple “pheromone-like” avoidance of previously visited sites. Simulate this model for an ensemble of $N = 1000$ independent walkers, each of which move $n = 500$ steps.

- (a) Plot the mean squared displacement as a function of number of steps. Compare this plot to the theoretical mean squared displacement for a 2D random walk without self-avoidance.
- (b) Fit a power law to your result in (a). To do so, first display the mean squared displacement as a function of step as a log-log plot. You should now be able to fit a linear function to this data and show that $\langle r^2(t) \rangle \sim t^\alpha$ for some α . Is the motion is subdiffusive ($\alpha < 1$), diffusive ($\alpha \approx 1$), or superdiffusive ($\alpha > 1$)?
- (c) In words, explain why self-avoidance can lead to faster spreading than simple diffusion at early times for a single walker.

We will now simulate multiple walkers with a shared avoidance strategy. You can think of this as a group of organisms (number of organisms is denoted N_{org}) that avoid their own and other's paths. When N_{org} walkers move simultaneously from the same origin, they share the visited set: any site visited by any walker counts as "visited" for all walkers. Update walkers in any fixed order each time step. Suppose these walkers move simultaneously from the same origin. Simulate $N_{org} \in \{1, 3, 10, 30, 100\}$ walkers starting at the origin, evolving for $T = 500$ steps under this model (all walkers avoid the union of visited sites). In each case, run $N_{runs} = 50$ independent realizations.

- (d) For each N_{org} , plot $\langle r^2(t) \rangle$ vs t .
- (e) Make a summary plot of the final-time mean squared displacement, i.e. plot $\langle r^2(T) \rangle$ as a function of N_{org} .